Section	Ask	Points	What good looks like	What average looks like	What poor looks like	What nothing looks like	
		60	**80-100%**	**60-80%**	**<60%**	**0**	**% Weightage**
Define the problem and perform an Exploratory Data Analysis	- Problem definition, questions to be answered - Data background and contents - Univariate analysis - Bivariate analysis	**8**	1) Definition of problem (as per given problem statement) and questions to be answered [1] 2) Observations on shape of data, data types of all the attributes, missing value detection, statistical summary [1] 3) Univariate Analysis (distribution plots of all the continuous variable(s), barplots/countplots of all the categorical variables) [3] 4) Bivariate Analysis (Relationships between important variables such as conversion status and time spent on the page, preferred language and time spent on the page, landing page type and time spent on the page) [3]	1) Definition of problem (as per given problem statements) 2) Observations on data types of various attributes, missing value detection, statistical summary. 3) Univariate Analysis (distribution plots of few continuous variables, barplots/countplots of few categorical variables) 4) Bivariate Analysis (Relationships between few unimportant variables)	1) Definition of problem (as per given problem statements) 2) Observations on data types of various attributes, statistical summary. 3) Univariate and Bivariate Analysis (any plot)		13.33
Illustrate the insights based on EDA	Key meaningful observations on individual variables and the relationship between variables	**6**	1) Comments on range of attributes, outliers of various attributes [2] 2) Comments on the distribution of the variables and relationship between variables [2] 3) Comments for each univariate and bivariate plots [2]	1) Comments on range of attributes, outliers of various attributes 2) Comments for some univariate and bivariate plots	1) A few random univariate done with little commentary 2) A few random bivariates done with little commentary		10.00
Prove(or disprove) that the medical claims made by the people who smoke are greater than those who don't?	"Perform the hypothesis test - Perform visual analysis - Formulate null and alternative hypotheses - Select the appropriate test - Data collection and preparation - Calculate the p-value - Write the inference based on the p-value"	**10**	1) Visual analysis (1) 2) Correct null and alternative hypothesis (2) 3) Selection of appropriate test (2) 4) Correct p-value using appropriate data and test parameters (3) 5) Appropriate conclusion and inference based on the p-value about the hypothesis(2)	1) Correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value	1) Partially correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value		16.67
Prove (or disprove) with statistical evidence that the BMI of females is different from that of males.	"Perform the hypothesis test - Perform visual analysis - Formulate null and alternative hypotheses - Select the appropriate test - Data collection and preparation - Calculate the p-value - Write the inference based on the p-value"	**10**	1) Visual analysis (1) 2) Correct null and alternative hypothesis (2) 3) Selection of appropriate test (2) 4) Correct p-value using appropriate data and test parameters (3) 5) Appropriate conclusion and inference based on the p-value about the hypothesis(2)	1) Correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value	1) Partially correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value		16.67

Does the smoking habit of customers depend on their region?	"Perform the hypothesis test - Perform visual analysis - Formulate null and alternative hypotheses - Select the appropriate test - Data collection and preparation - Calculate the p-value - Write the inference based on the p-value"	**10**	1) Visual analysis (1) 2) Correct null and alternative hypothesis (2) 3) Selection of appropriate test (2) 4) Correct p-value using appropriate data and test parameters (3) 5) Appropriate conclusion and inference based on the p-value about the hypothesis(2)	1) Correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value	1) Partially correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value		16.67
Is the mean BMI of women with no children, one child, and two children the same? Explain your answer with statistical evidence.	"Perform the hypothesis test - Perform visual analysis - Formulate null and alternative hypotheses - Select the appropriate test - Data collection and preparation - Calculate the p-value - Write the inference based on the p-value"	**10**	1) Visual analysis (1) 2) Correct null and alternative hypothesis (2) 3) Selection of appropriate test (2) 4) Correct p-value using appropriate data and test parameters (3) 5) Appropriate conclusion and inference based on the p-value about the hypothesis(2)	1) Correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value	1) Partially correct null and alternative hypothesis 2) Selection of appropriate test 3) Partial data collection 4) Incorrect p-value 5) Appropriate conclusion and inference based on the p-value		16.67
Presentation/Notebook - Overall quality	- Structure and flow - Crispness - Visual appeal - Conclusion and Business Recommendation OR - Structure and flow - Well commented code - Conclusion and Business Recommendation	**6**	- Clear structure and flow - everything sits well in a story - (Problem - Data overview - Solution overview - Findings - Recommendations) [1] - Crispness - not too many words - just enough to keep the focus on key things/points [1] - Visual appeal - use of charts, colors, diagrams, format, symmetry - informative visualizations that are easy to interpret [2] - Conclusion and Business Recommendation based on the analysis. [2] OR -Well structured notebook with a logical flow [2] -Clean and well commented code [2] - Conclusion and Business Recommendation based on the analysis. [2]	- There is structure and flow but some bits are missing / jumbled - Points are made but in too many words - lesser charts, format is not the cleanest - Conclusion and recommendations are missing OR - There is structure and flow but some bits are missing - Some of the code is commented - Conclusion and recommendations are missing	- no structure or flow - only a few points are covered - story is not complete - not many visuals used - Conclusion and recommendations are missing OR - no structure or flow - no comments in the code - Conclusion and recommendations are missing	- no presentation OR - no code	10.00

Section(s)	Important Notes for the Evaluators
Overall Submission	- Learners have an option to choose between notebook submission or presentation submission. - Grading for all sections except Presentation/Notebook Overall Quality will be the same for both types of submission. - In Presentation/Notebook Overall Quality, rubric comments are separated for Presentation and Notebook using the word OR
Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?	- For this problem, Two proportions z-test/Chi-square test of independence/Fishers' exact test can be used to draw the inference. - We have included the two proportions z-test in our sample solution. But, the learners may use either of the three tests to draw the conclusion. The p-value might be different depending on the test. Marks should not be deducted in that case.
All the sections where Hypothesis Testing has been used.	- Please follow the weightage given in this rubric to deduct marks for the hypothesis testing. - Some learners may use non-parametric tests (advanced tests not covered in the course) to answer the questions. In that case, marks should not be deducted if the approach is correct. - Marks should not be deducted if the learner uses the Rejection Region approach to perform hypothesis testing instead of the p-value approach.